③ A : {N, N+2, N+4, N+7, N+10, N+13, N+15}
: {18, 20, 22, 25, 28, 31, 33}

B : {N+3, N+5, N+7, N+10, N+12, N+18}
: {21, 23, 25, 28, 30, 36}

C : {N+1, N+7, N+10, N+11, N+17, N+20}
: {19, 25, 28, 29, 35, 38}

Universe U : {18, 19, 20, 21, 22, 23, 25, 28, 29, 30, 31, 33, 35, 36, 38}

a) |A ∪ B ∪ C| : {18, 19, 20, 21, 22, 23, 25, 28, 29, 30, 31, 33, 35, 36, 38}
: 15

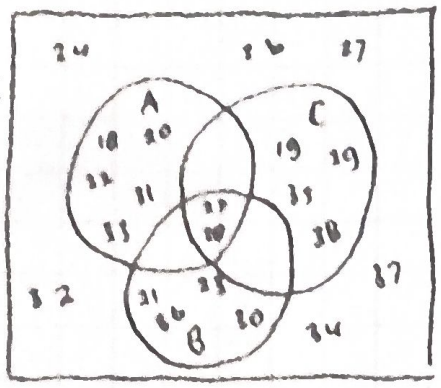
|A ∩ B ∩ C| : {25, 28}
: 2

|(A ∪ B) ∩ C'| -> • A ∪ B : {18, 20, 21, 22, 23, 25, 28, 30, 31, 33, 36}
• C' : {19, 25, 28, 29, 35, 38}
C' : {18, 20, 21, 22, 23, 24, 26, 27, 30, 31, 32, 33, 34, 36, 37}

|(A ∪ B) ∩ C' -> {18, 20, 21, 22, 23, 30, 31, 33, 36}
: 9

A ⊕ B : (A ∪ B) - (A ∩ B)
: {25, 28} - {18, 20, 21, 22, 23, 29, 28, 30, 31, 33, 36}
A ⊕ B : {18, 20, 21, 22, 23, 30, 31, 33, 36}
: 9

diagram venn :



c) Dik :

$f(x) : k \cdot x + (a + b)$

- $f(x) : 10x + 9$

$g(x) : \dfrac{(x^2)\ f'}{(b+1)+a}$

- $g(x) : \dfrac{x^2}{9} + 1$

a. $(f \circ g)(x) \rightarrow f(g(x)) = 10\left(\dfrac{x^2}{9} + 1\right) + 9$

$= \dfrac{10x^2}{9} + 10 + 9$

$= \dfrac{10x^2}{9} + 19$

- $(g \circ f)(x) \Rightarrow \dfrac{((10x+9)^2)}{9} + 1 \rightarrow (10x+9)^2 = 100x^2 + 180x + 81$

$= \dfrac{(100x^2 + 180x + 81)}{9} + 1$

$= \dfrac{100x^2 + 180x + 81 + 9}{9}$

$= \dfrac{100x^2 + 180x + 90}{9}$

a apakah komutatif? tidak karena

- $(f \circ g) \Rightarrow : \dfrac{10x^2}{9+19}$
- $(g \circ f) : \dfrac{100x^2}{9 + 20x + 10}$

uji $x = b + 2$
$= 10$

$\rightarrow f(g(10)) : 10\left(\dfrac{100}{9} + 1\right) + 9$

$= 10\left(\dfrac{100}{9} + \dfrac{9}{9}\right) + 9$

$= 10\left(\dfrac{109}{9}\right) + 9$

$= \dfrac{1090}{9} + 9$

ubah $9 \rightarrow : \dfrac{1090}{9} + \dfrac{81}{9}$

$= \dfrac{1171}{9}$

$g(f(10)) = \dfrac{(109)^2}{9} + 1$

$= \dfrac{11881}{9} + 1$

$= \dfrac{11881}{9} + \dfrac{9}{9}$

$= \dfrac{11890}{9}$ $\rightarrow$ kesimpulan $\dfrac{1171}{9} \neq \dfrac{11890}{9}$ (tidak komutatif)

d) invers $f^{-1}(x)$

$f(x) = 10x + 9$

$y = 10x + 9$

$x = 10y + 9$

$x - 9 = 10y$

$y = \frac{x-9}{10}$

$f^{-1}(x) = \frac{x-9}{10}$

· volume pembelian x → $= x \cdot 100 + a + b$

$= 10 \cdot 100 + 1 + 8$

$= 1000 + 9$

$= 1009$

substitusi ke invers → $x = f^{-1}(1009)$

$= \frac{1009 - 9}{10}$

$= \frac{1000}{10}$

volume pembelian x = 100

verif → $f(100) = 10(100) + 9$

$= 1000 + 9$

$= 1009$ ✓

e. $f(x) = 10x + 9$

· ingektif → jika $f(x_1) = f(x_2)$, maka harus terbukti $x_1 = x_2$

$10x_1 + 9 = 10x_2 + 9$

$10x_1 = 10x_2$

$x_1 = x_2$ → ingektif ✓

· surgektif → harus bisa menghasilkan semua $y \in A$

misal $y = 10x + 9$

$x = \frac{y-9}{10}$ → surgektif ✓

karena untuk setiap y ada x di $\mathbb{R}$.

bijektif ✓

$g(x) = \frac{x^2}{9} + 1$

· ingektif

$g(3) = \frac{9}{9} + 1 = 2$

$g(-3) = \frac{9}{9} + 1 = 2$

} $3 \neq -3$ → tidak ingektif

· surgektif

$g(x) = \frac{x^2}{9} + 1 \geq 1$ → nilai terkecil $= 1$

tidak bisa menghasilkan nilai < 1

padahal kodomain $= \mathbb{R}$

→ tidak surgektif

→ tidak bijektif